

What a Two-Byte Witness Buys on a Sprayed Fabric:

A Silicon-Measured Comparison of In-Fabric Grayhole Localization Against Passive Detection

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Abstract—Packet spraying spreads each flow of a machine-learning training job across every spine switch, so a link that silently drops a small fraction of its packets leaves its evidence divided among many paths. A detector that only counts arrivals must separate a shrinking deficit from spray noise, at a cost growing as the inverse square of the loss rate, and one vantage cannot tell an uplink from the downlink behind it. In this paper, we measure what an in-fabric per-directed-link witness buys against the two most recent passive detectors and against a zero-byte counter pair that holds the same information. The witness is an existing primitive: two bytes per packet and two registers per directed link. We replay all three comparators over one shared stream and measure the ledger on an Intel Tofino switch. The witness detects on one epoch of two million packets from a loss rate of 1.5 percent down to one in ten thousand, where the passive cost has grown to ninety-four million packets and then past the budget; its own wall sits at a few times the background loss floor and moves with it. It names the exact directed link, by construction given per-link counts, where the passive detectors return the uplink and downlink pair. A counter pair matches it only when both registers are read within about ten microseconds; at the 2.6 milliseconds we measured it false-alarms on every trial. The advantage fails under a fleet-wide shift and under incast, where a relative passive test is better.

Index Terms—Data center networks, packet spraying, gray failure, network telemetry, programmable switches.

I. INTRODUCTION

Training fabrics for large machine-learning models spray packets. Instead of pinning a flow to one path, the leaf switch sends consecutive packets of the same flow to different spines, which balances load across a Clos fabric and is now the default of the Ultra Ethernet transport [1]–[3]. Spraying also changes what a failure looks like. A link that drops one packet in a thousand, or one in ten thousand, does not take a flow down. It slows every flow a little, because every flow crosses it. Such gray failures are common in production fabrics and hard to attribute [4], [5], and they can be costly for synchronous training jobs whose collective operations wait for the slowest packet [6]. We call a link that silently drops a small fraction of its packets a grayhole, and we call the problem of naming the directed link that carries it grayhole localization.

Grayhole localization on a sprayed fabric is challenging, due to two reasons. First, the evidence is divided. A flow that used to concentrate its loss on one path now spreads it across k

spines. The per-path deficit a passive observer can see is then $1/k$ of the fault, and it shrinks further as the loss rate falls, while the sampling noise of spraying, i.e., the variation in how many packets each spine happens to receive, does not shrink at all. Second, the evidence is aliased. A packet from one leaf to another crosses two directed links, an uplink into a spine and a downlink out of it. An arrival counter at the destination sees only the product of the two survival rates, and thus, a single vantage that counts arrivals can name a spine but cannot, by itself, tell the uplink from the downlink.

The two most recent detectors built for sprayed fabrics accept both limits. SprayCheck [7] applies a Z-test to per-spine arrival counts and resolves a directed link by intersecting reports from several leaves. FlowPulse [8] compares each ingress port's load against a learned baseline and applies a per-sender rule. Both are passive: they add nothing to the wire, they run on counters the switch already has, and they can be deployed today. However, a test on arrival counts needs enough packets for a deficit of fraction p to stand above spray noise, and that number grows at least as $1/p$. Both papers also state that one report can only flag the uplink and downlink pair [7], [8].

An in-fabric witness can remove both limits by construction. The sending leaf stamps every packet with a per-directed-link sequence number. The receiving switch keeps, per directed link, the highest sequence number seen and the number of packets seen. The difference of the two deltas is then the loss on that directed link and nothing else. This primitive is not new. NetSeer [9], LinkGuardian [10], LossRadar [11], and the link-layer retry of Ultra Ethernet [2] all place a per-link sequence witness in the data plane. Nor is the information new: the difference is a transmit count minus a receive count, and every switch keeps per-link counters of both at zero wire cost. What none of the prior systems reports is what the witness buys, and costs, on a sprayed fabric at the loss rates where the passive state of the art fails, and what separates it from the counters the switch already has. That measurement, head-to-head against the passive detectors and against a counter pair, and on real silicon, is the contribution of this paper. We implement the witness as a receiver ledger on an Intel Tofino switch [12], with a two-byte stamp and two registers per directed link, and we treat it as the measured arm of a comparison rather than as a new design.

To make the comparison fair, we replay SprayCheck and

FlowPulse from their own papers, with their own thresholds and their own localization rules, and we give the counter pair the ledger's own information and decision rule, perturbed only by the time between its two register reads. All four arms read one shared stream of sprayed traffic, and each sees it only through the counters its own switch would have. We run fifty seeds per cell from 1.5 percent loss down to the background floor, and we report false positives next to every action rate. Where a comparator ties or beats the witness, we report it as plainly as where it does not. Our contributions are as follows.

- **States and measures the detection-cost separation.** A count test and a witness both need packets that grow as the inverse square of the loss rate; the witness divides the constant by the background floor (Proposition 1). Measured from the fault's onset, the witness detects on one epoch of 2 million packets from 1.5 percent down to 10^{-4} with no false positive. SprayCheck's median cost has grown to 94 million packets at 10^{-3} , 47 times the witness, with an action rate of 0.78, and it detects nothing within a 160-million-packet budget at 10^{-4} ; FlowPulse fails one order earlier. The witness's own wall appears at a few times the floor, at 6 and 36 million packets at 5×10^{-5} and 2×10^{-5} over a 10^{-5} floor. A sweep over three floors shows the wall moving with the floor (Section V).
- **Measures what the two bytes buy against a zero-byte counter pair.** With its two registers read at one instant the counter pair reproduces the witness cell for cell. With a read skew of ten microseconds against a 100 ms epoch it false-alarms on a healthy link in 28 to 100 percent of trials. At the 2.6 ms pairwise read and the 350 ms census we measured on our controller it false-alarms in every trial and never localizes. What the stamp buys is in-band alignment of the two counts (Sections V and IX).
- **Measures how far a passive vantage can disalias under spraying.** With the witness as the oracle row, SprayCheck resolves the exact directed link at 1.0 to 1.5 percent loss, which we report as a null result, and degrades to the ambiguous uplink and downlink pair at 10^{-3} . FlowPulse cannot detect an uplink fault at any rate once several senders share the port (Section VI).
- **Measures the ledger on silicon.** On a Tofino switch the ledger recovers injected loss exactly from 10^{-2} down to 10^{-4} , i.e., six drops in sixty thousand packets, with zero false positives on more than forty thousand clean packets, for scheduled and for random drops. It attributes a downlink grayhole to the downlink alone, with zero false attribution on seven active sublinks (Section VII).
- **Bounds the claim.** Under two or three simultaneous independent faults the witness's advantage widens, because the passive intersection produces spurious links. Under a fleet-wide shift in loss with no culprit, and under persistent congestion loss on one leaf's eight downlinks, the witness names the shared loss as faults and SprayCheck's relative test is the better tool. We report all three, and we scope every other claim to independent per-directed-link faults (Section VIII).

- **Quantifies the cost, including two costs the reduction to two bytes introduced.** The witness adds 0.14 to 2.4 percent to the wire depending on packet size, six bytes of register state per directed link, and one ingress pipeline stage, which brings the program to the twelve-stage ceiling of the Tofino 1. The reduction also introduced an aliasing of unsprayed traffic and a rare, still unexplained one-packet anomaly that the four-byte program did not show (Sections IX and X).

To the best of our knowledge, this is the first head-to-head measurement of an in-fabric per-directed-link witness against the passive sprayed-fabric detectors and against a zero-byte counter pair, on the loss regime all of them target, and on silicon. We claim no novelty for the witness itself. The per-link sequence stamp is the primitive of NetSeer and LinkGuardian, and the localization advantage we measure is a property of the information the witness holds rather than of a better inference. We also report a measurement that weakens the motivation as it is usually stated. In a packet-level simulation, a 10^{-3} grayhole on a fully loaded uplink slows a collective by 1.4 percent under timeout-dominated recovery, so the value of naming such a link lies in repair and attribution rather than in the running job (Section X). What the paper adds is the measurement, its cost, and its boundary.

Sections II to IV give the fault model, the ledger, and the harness; Sections V to IX report the four objectives and the cost; Sections X to XII discuss limitations, position the work, and conclude.

II. BACKGROUND, FAULT MODEL, AND OBJECTIVES

A. Sprayed Leaf-Spine Fabrics

A two-tier Clos fabric connects n leaf switches to k spine switches, and every leaf has one link to every spine. A packet from leaf L_a to leaf L_b crosses exactly two directed links: the uplink $L_a \rightarrow S_i$ and the downlink $S_i \rightarrow L_b$ for some spine S_i . We call each such one-way hop a *directed link*. A fabric with n leaves and k spines has $2nk$ of them. Under per-packet spraying the leaf chooses the spine for each packet independently, in the simplest case uniformly at random [1]. A flow of N packets then places about N/k packets on each of its k uplinks and on each of its k downlinks. The Ultra Ethernet specification adopts spraying as its default and adds a link-layer retry that can retransmit a lost frame across one link [2]. REPS extends spraying with a load balancer that steers away from paths that show loss [3]. Throughout the paper we use $n = 4$ and $k = 8$ in the replay harness, which is the topology SprayCheck evaluates [7], and a 4-leaf, 2-spine virtual fabric on silicon.

A *grayhole* is a directed link that forwards most of its packets and silently drops a fraction p of them. We study p from 1.5 percent down to 10^{-4} , and we call 10^{-3} to 10^{-4} the low-loss regime. A grayhole differs from a blackhole, which drops everything and is caught by any liveness check. It also differs from congestion loss, which is transient and is shared by every link behind the congested queue. Time is divided into *epochs*, i.e., the period at which a controller reads counters and decides. In the replay harness one epoch carries

two million packets across the fleet, and the detectors act at epoch boundaries.

B. Two Limits of a Passive Vantage

A passive detector counts packets the switch already forwards and adds nothing to them. Under spraying it faces two limits, and both become worse as p falls.

The cost limit. A grayhole of rate p on one of k spines removes a fraction p of the packets that spine should have received. To notice the deficit, a passive detector compares that spine's arrival count against the others, and the comparison is only meaningful when the expected deficit exceeds the sampling noise of spraying. A witness that observes each directed link's own transmit count compares the link's loss count against the background loss floor f instead. Both comparisons are Poisson separations, and both scale the same way in p ; what differs is the constant. We state this as a proposition, because it governs every detection number in the paper.

Proposition 1 (Detection cost of a count test and of a witness). *Let a directed link carry λ packets per epoch, let spray decisions and per-packet drops be independent, and let a test require s standard deviations of separation. A passive test on arrival counts separates a loss fraction p from spray noise, which has standard deviation $\sqrt{\lambda}$, when $\lambda p > s\sqrt{\lambda}$, i.e., when*

$$\lambda_{\text{passive}} > s^2/p^2. \quad (1)$$

A witness that counts the link's own drops separates them from a background floor $f < p$, whose drops have standard deviation $\sqrt{\lambda f}$, when $\lambda p > s\sqrt{\lambda f}$, i.e., when

$$\lambda_{\text{witness}} > s^2 f/p^2. \quad (2)$$

The two bounds have the same $1/p^2$ form; the witness divides the constant by f , which is 10^{-5} on a fabric whose floor is 10^{-5} . The Gaussian form of (2) holds when the floor produces many drops per epoch; when λf is below about one, the separation is a sparse Poisson test and the required λ grows as $1/(p \log(p/f))$ instead, which is milder in p but diverges as p approaches f . Two consequences follow, and Section V measures both. First, over any range of p for which one epoch's λ already exceeds $s^2 f/p^2$, the witness detects on the first epoch and its cost looks flat; that is a statement about the epoch volume relative to the floor, not about the absence of a wall. Second, the witness has a wall of its own, at p of a few times f , and the wall moves with f .

The aliasing limit. An arrival counter at L_b for packets that came through spine S_i sees the product of the survival rates of $L_a \rightarrow S_i$ and $S_i \rightarrow L_b$. One counter alone cannot tell which of the two directed links dropped the packets. SprayCheck resolves the pair by intersecting failure reports from several leaves that share the spine [7]. FlowPulse resolves it by asking whether every sender on the port is affected or only one [8]. Both rules need corroborating traffic, and that traffic can become scarce at low loss. A per-directed-link witness holds each link's own count and needs no corroboration. Section VI measures both.

C. Fault Model, Objectives, and Scope

We assume that the loss rate of a faulty directed link is stationary over the observation window and that faults on different directed links are independent. We argue that this is a reasonable assumption for the physical-layer causes that dominate gray failures in production, i.e., damaged optics, marginal cables, and misconfigured forward error correction. Each of these is local to one link and persists until repaired [5]. We assume that a healthy directed link has a small background loss rate, set to 10^{-5} in the harness, and we assume that one or several directed links can be faulty at once. We also assume that congestion loss is negligible on the measurement timescale, as on a lossless fabric with priority flow control or one provisioned so that queue drops are rare; this is a precondition of the absolute-floor test rather than a convenience, because the test counts a queue's drops exactly as it counts a link's, and Section VIII shows what happens when the assumption fails. We do not assume that the faulty link is known to carry any particular flow, and we do not assume that the controller can inject probes.

The paper evaluates four objectives, and we refer to them by label in the rest of the paper.

- **O1, detection cost.** The number of packets, and the fraction of trials, a detector needs to flag the faulty spine or link within a fixed budget, as a function of p , with the false-positive rate reported next to it.
- **O2, localization.** Whether the detector names exactly the faulty directed link, an ambiguous set that contains it, or nothing.
- **O3, silicon fidelity.** Whether the ledger, on a real switch, recovers the injected loss exactly at each p and attributes it to the injected directed link alone.
- **O4, the boundary.** Whether O1 and O2 survive several simultaneous faults, and whether they survive a fleet-wide, correlated shift in loss.

Scope. Two regimes are out of scope for the headline claims. Under a common-mode shift, in which every link's loss rises together, the ledger's decision rule fails, as Section VIII shows and Section X discusses. The claims of Sections V to VII are therefore stated for independent faults on a stationary fabric. Repairing or re-admitting a faulty link is not addressed; Section XI cites the systems that do. Blackholes are used only to validate the testbed and are not part of the evaluation.

III. THE RECEIVER LEDGER

The receiver ledger is the measured arm of this paper. It is our implementation of the per-link sequence witness that NetSeer [9] and LinkGuardian [10] place in the data plane, at reduced width, and we describe it here far enough that the cost, the decision rule, and the measurements of the later sections can be read and reproduced. We claim no design contribution for it. In Figure 1, we show the data path from a sending leaf to a receiving leaf and the two places the ledger touches it.

A. Data Plane

Witness stamp. At the egress of the sending hop, the switch inserts a two-byte header carrying a 16-bit sequence number

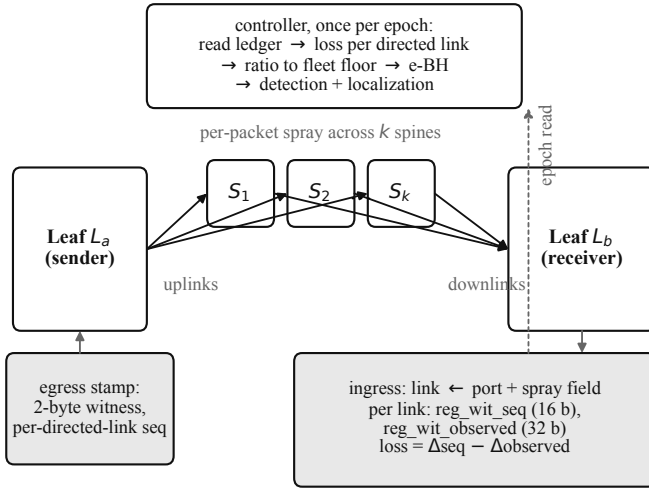


Fig. 1. The receiver ledger on a sprayed leaf-spine fabric. The sending leaf stamps each packet at egress with a two-byte per-directed-link sequence number. The receiving switch reconstructs the directed link from its own ingress port and the spray field, and keeps two registers per directed link: the last sequence number seen and the number of packets seen. Once per epoch the controller reads both registers, takes the difference of their deltas as the loss on that directed link, compares it with the fleet floor, and applies e-BH to name the faulty links.

that it increments per directed sublink, i.e., per directed link and traffic class. The header carries nothing else. An earlier version of the ledger also carried a 16-bit link identifier, which doubled the wire cost. Because the receiving hop already knows its own ingress port, and because the spray field is already on the wire for load balancing, the identifier is redundant for sprayed traffic, and we removed it. Section IX reports what that removal cost in pipeline stages, and Section X reports a correctness cost it introduced for traffic that carries no real spray choice. The stamp is an in-fabric cost only: it is stripped before the packet reaches a host.

Link reconstruction. At the ingress of the receiving hop, a match table keyed on the ingress port and the spray field names the directed link, and a second table appends the traffic class, which the switch classifies fresh for every packet from the packet's size and its differentiated-services code point. Two tables are needed rather than one because the traffic class is a per-packet property, not a per-link one; a single table that baked a fixed class into each link entry would mislabel every packet whose class differed from the entry.

Ledger registers. For each of the $2nk$ directed links, and for each traffic class, the receiving switch keeps two registers: *seq*, the 16-bit sequence number of the last stamped packet seen on that sublink, and *obs*, a 32-bit lifetime count of the stamped packets seen. Both are updated in the data plane on every stamped packet. Over an epoch in which the two registers advance by Δseq , taken modulo 2^{16} , and Δobs , the loss on the sublink is

$$\ell = \Delta seq - \Delta obs, \quad (3)$$

because the sender advanced the sequence once per packet it sent and the receiver counted once per packet that arrived. Equation (3) is a transmit-minus-receive difference; the sequence number is not consumed as an ordering, and what

the in-band stamp adds over a pair of counters read on two switches is that both terms are read on one switch in one operation (Section IV defines the counter-pair arm that isolates this).

What the identity does and does not see. Because the sequence register keeps a frontier rather than a running sum, a late packet manufactures no phantom loss; within one sublink, i.e., one link and one queue, packets are not reordered in the first place, so this property matters only across the epoch boundary. A packet lost after the last survivor of an epoch is counted in the next epoch, when the next survivor advances the frontier. A duplicated packet drives the estimate to -1 , which the controller reads as a duplicate; a duplicate and a loss in the same epoch on the same sublink cancel to zero, so the signature survives only when duplicates are unaccompanied. A sublink that drops every packet in an epoch advances neither register, and Equation (3) reads zero: total failure is invisible to the ledger by construction and needs a liveness companion, which is why blackholes are used only to validate the testbed in this paper. We verified the frontier, duplicate, and trailing-loss behaviours against the compiled program on the vendor's software model before any hardware measurement (Section IV).

B. Decision Rule

Once per epoch the controller reads every sublink's registers and tests each sublink on its own evidence against a floor estimated from the rest of the fleet. The rule has four parts, and we state each with the values used in every experiment of this paper.

Floor. The floor f_t for a sublink at epoch t is the leave-one-out loss rate of the fleet's currently healthy sublinks over the last $W = 20$ epochs. It excludes the sublink under test and every sublink currently rejected, so that a faulty link cannot inflate the floor used to judge it or its siblings. When the leave-one-out pool carries too little traffic to trust, the epoch is censored for that sublink rather than judged against a fallback constant; a fixed fallback floor was tried and produced false alarms on every epoch of a healthy single-sublink fleet.

Evidence. For a sublink that transmitted λ packets and lost ℓ of them in epoch t , the per-epoch evidence is a convex mixture of likelihood ratios of the binomial loss count under alternatives $p_1 = rf_t$ against the null $p_0 = f_t$, over the ratio grid $r \in \{2, 5, 10, 50, 200\}$:

$$E_t = \frac{1}{|R|} \sum_{r \in R} \frac{\Pr[\ell \mid \lambda, rf_t]}{\Pr[\ell \mid \lambda, f_t]}. \quad (4)$$

A fixed mixture of likelihood ratios is an e-variable, and its running product over epochs is an e-process, because the null f_t is previsible: it is estimated without the sublink's own current-epoch outcome, so the expectation of each factor under the true null is one whatever the null's value that epoch. A censored epoch contributes a factor of exactly one, with no reset and no alpha spending. The wealth $M_t = \prod_{s \leq t} E_s$ is reset only when the link is repaired, since a repaired link has a new loss rate by construction.

Multiplicity. The fleet decision at epoch t applies the e-BH procedure of Wang and Ramdas [13] to the vector of wealths

M_t over all sublinks at a false discovery rate of $\alpha = 0.05$. Sublinks are ordered by wealth, and the largest k^* such that the k^* -th largest wealth is at least $|S|/(k^*\alpha)$ are rejected, where $|S|$ is the number of tested sublinks. e-BH controls the false discovery rate under arbitrary dependence, which matters here because sublinks share spines and the floor. The rejected set is both the detection and the localization: a rejected sublink is a directed link.

Reported false positives versus α . The evaluation reports a false-positive rate of 0.00 in every single-fault cell. That is not α driving the rate to zero; it is the e-BH threshold being inert when the faulty link's wealth is large and every healthy link's wealth is near one, which is the regime in which the fleet floor is well estimated. The regime in which the floor is not well estimated is exactly the one in which the rule fails (Section VIII): under a fleet-wide shift the floor moves with the fault, healthy links accumulate wealth against a lagging null, and e-BH rejects nearly all of them.

IV. METHODOLOGY

The evaluation has two vehicles. A replay harness compares the ledger with the two passive detectors on one shared stream of sprayed traffic, because the passive detectors are host-side systems that cannot run on our fabric and because a head-to-head needs every arm to see the same packets. A silicon testbed then measures the ledger's own side of that comparison on a real switch.

A. The Replay Harness

Shared stream. Every epoch, the harness draws one per-packet spray decision and one per-hop survival outcome for every packet, and it feeds the same draw to all four arms. Spray is independent and uniform across the $k = 8$ spines, which matches SprayCheck's own model and FlowPulse's temporal-symmetry assumption [7], [8]. Survival is independent per directed link with rate $1 - p$ on a faulty link and $1 - 10^{-5}$ on a healthy one. Each arm sees only what its own switch would see. The ledger arm receives the exact per-sublink transmit and receive counts its registers produce. The SprayCheck arm receives receive-only per-spine arrival counts; no transmit count and no drop count exists in its interface, so none can be passed to it by mistake. The FlowPulse arm receives receive-only per-port load, predicted from its own bootstrapped history, and never the ledger's transmit ground truth.

Scenario. Each trial runs ten epochs of clean traffic, during which FlowPulse bootstraps its baseline and the ledger warms its floor. The trial then degrades one spine, or one directed link, to the target rate p for the rest of the run. For detection (O1) the fleet carries two million packets per epoch, and a trial ends when the arm flags the faulty spine or after 80 post-onset epochs, a budget of 160 million packets. For localization (O2) the harness extends to the two-hop fabric of $n = 4$ leaves, and each ordered leaf pair carries two million packets per epoch. The localization budget is 60 post-onset epochs, which is deliberately generous to the passive arms so that SprayCheck can complete a corroborating round of measured flows. For the

correlated gate (O4) the window is 40 post-onset epochs. We run 50 seeds per cell.

Two fidelity checks that changed the harness. Before trusting any number we checked each arm against its own paper. Two checks failed and were fixed. First, a per-packet traffic generator made one two-million-packet epoch take tens of seconds; we vectorized it with a multinomial draw, which changed no distribution. Second, at an initial volume of 200 thousand packets per epoch, FlowPulse's fixed one-percent threshold false-alarmed on a healthy spine in every trial. The reason is that the relative spray noise at that volume, about 0.6 percent, sat close to its threshold. We raised the volume to two million packets per epoch, at which its false-positive rate on healthy spines is 0.0000, consistent with the paper's own claim of under one percent [8]. We report that rate next to every result.

Pre-registration and artifact. The claim boundary of this paper was pre-registered before the correlated-fault gate of Section VIII was run: the pre-registration's version 1.9 restricts the headline claims to independent per-directed-link faults, records common-mode degradation as a disclosed limitation, and defers its fix as an untested hypothesis. The same record holds six earlier novelty gates, each with an adverse verdict, which is why the paper claims no novelty for the primitive. The replay harness with its seeds, both baseline reimplementations, the ledger's P4 source, both compiler resource reports, the controller and e-BH code, and the raw silicon census logs, including the soak that produced the anomaly of Section X, are in the public repository named in the acknowledgment; every number in the paper is listed there with its source file.

B. The Four Arms

The ledger. The controller of Section III, unmodified from the version measured on silicon, runs over the exact per-sublink counts. Its localized set is the set of sublinks e-BH rejects.

SprayCheck-Z. We reimplement SprayCheck's detector from its paper [7]: a Z-test on per-spine arrival counts with sensitivity s calibrated under the paper's own Poisson model at 2.5 million packets per spine, which gives $s = 3.24$. For localization we implement its full cross-leaf rule: a report flags the uplink and downlink pair of a path, and a single link is named only when it lies in the intersection of reports from different leaves. SprayCheck's real spraying uses join-the-shortest-of-two-queues, which has less noise than the Poisson model; a sensitivity tuned to that would shift its cost curve down by a constant factor but not change how the cost scales with p .

FlowPulse- θ . We reimplement FlowPulse's port-load detector with its fixed one-percent threshold against a learned per-port baseline, and its per-sender localization rule: if every sender on an ingress port is affected the local downlink is named, and if only one sender is affected the uplink from that sender's leaf is named [8]. We give the arm per-sender visibility on every port, which its paper assumes; this is the charitable reading, not a weakened one.

CounterPair-OB. The fourth arm answers the question a reviewer asks first: every switch already has per-link transmit

and receive counters at zero wire cost, so what do the two bytes buy? Equation (3) is a transmit-minus-receive difference. A counter pair keeps the transmit register on the sending switch and the receive register on the receiving switch, with no stamp, and the controller reads them on two devices at two instants. We give this arm the same per-directed-link information as the ledger and the same decision rule, and we perturb only its transmit reading. Let the transmit read of epoch e land d_e^{tx} after the epoch boundary and the receive read d_e^{rx} after it, each drawn independently and uniformly from $[0, \sigma)$, where σ is the span of the controller's read loop as a fraction of the epoch. The observed transmit delta is then the true delta plus $\lambda(o_e - o_{e-1})$ with $o_e = d_e^{tx} - d_e^{rx}$, a zero-mean disturbance with standard deviation near $\lambda\sigma/\sqrt{6}$ that the arm cannot tell from loss. Packets in flight on the link are microseconds and are not modelled. If the disturbance would put transmit below receive the arm clamps to zero loss, as a controller would. At $\sigma = 0$ the arm is the ledger's information read at one instant, and it is the idealized bound. We give the arm the ledger's own absolute decision rule deliberately, to isolate the information difference; a skew-aware counter pair that normalized a leaf's uplink registers against each other, whose read offsets are correlated because they sit on one switch, would cancel part of the common skew and is the stronger baseline we did not build. We sweep σ from 0 to 0.1, and we anchor it with a measurement. On our switch a pairwise read of one sublink's two registers takes 2.6 ms and a census of all 1024 sublinks takes 350 ms (Section IX). Against the harness's 100 ms epoch the best case for a counter pair is therefore $\sigma = 0.026$, and the full-census case is a sweep longer than the epoch itself.

C. Metrics

Action rate is the fraction of the 50 seeds in which the arm flags the faulty spine or link within the budget, with a 95 percent Wilson interval [14]. **Packets to detect** is the fleet-wide packet count from the fault's onset to the first correct flag, reported as a median with its interquartile range over the seeds that detected; we mark it as survivorship-biased wherever the action rate is below one. **False positive** is any healthy spine or link flagged, reported next to every action rate. For O2, **exact** means the arm named the true directed link and only it, **ambiguous** means it named a set that contains a healthy link, and **miss** means it named nothing. We also report the mean set size over detections and a paired exact McNemar test between arms on the same seeds. For O4, **recall** is the fraction of true faulty links named, and **exact-all** is the fraction of seeds that named the true set and nothing else. We report the false-positive rate both over the whole window and at its final epoch, so that a transient false flag can be separated from a persistent one.

D. The Silicon Testbed

Switch. The ledger runs on an Intel Tofino 1 switch [12] with SDE 9.13.2. A 4-leaf, 2-spine virtual fabric is emulated on the switch itself: a four-lane 25 Gb/s direct-attach cable loops four front-panel ports back to the switch, and each directed virtual link is one traffic-manager queue on one

loop port. With four traffic classes, the program tracks 1024 sublinks. The pipeline occupies 12 ingress and 5 egress match-action stages (Section IX).

Traffic and ground truth. A server with a 25 Gb/s network interface sends 1400-byte packets at 20 thousand packets per second, and a differentiated-services code point selects the traffic class and thus the sublink. Two fault injectors run in the switch's egress pipeline. The exact injector drops exactly D stamped packets on a chosen sublink, so the ground truth is D . The stochastic injector draws a 16-bit random value per packet and drops the packet when the value falls below $W = \lfloor p \cdot 2^{16} \rfloor$, so it drops each packet with probability $W/2^{16}$. For this injector the ground truth is its own direct counter of packets dropped, read from the switch, because a random injector has no preset count. Every cell reads a baseline census immediately before arming and reports deltas against it. Every cell also keeps Δ_{seq} below 2^{16} , so that the sequence wraps at most once, and clears both injectors before and after. The recovered loss is Equation (3) read through the controller's census.

Model validation. Before any hardware run, the compiled program passed nine correctness tests on the vendor's software model. They cover the exact recovery of injected loss, the invisibility of a trailing loss until the next survivor, the 32-bit transmit frontier past 255 and 256 packets, the absence of phantom loss under reordering, the -1 signature of a duplicate, and the realized rate of the stochastic injector, which fell within four standard deviations of its target. The hardware measurements of Section VII were then taken on the same build.

V. DETECTION COST (O1)

Proposition 1 predicts three things: the passive cost grows as $1/p^2$; the witness's cost is set by the epoch volume until p approaches the floor f , where it has a wall of its own; and a zero-byte counter pair with the same information as the witness is separated from it only by read skew. This section measures all three. Packets to detect are counted from the fault's onset. In Figure 2 and Table I, we show the packets each arm needed to flag the faulty spine and the fraction of seeds in which it did so, for eight loss rates from 1.5 percent down to the floor of 10^{-5} .

The ledger detects on one epoch until the floor is near. From 1.5 percent down to 10^{-4} the ledger flagged the faulty spine in every seed on the first epoch after onset, at a cost of 2 million fleet packets with an interquartile range of 2 to 2 million, widening to 2 to 4 million at 10^{-4} . The reason is that one epoch carries 250 thousand packets per spine, which exceeds the $s^2 f/p^2$ of Proposition 1 at every rate in that range, so the first epoch's drops can already stand above the floor. Below 10^{-4} the wall appears where the proposition puts it. At 5×10^{-5} the median cost rose to 6 million packets and at 2×10^{-5} to 36 million, still with an action rate of 1.00 and no false positive. At the floor itself, 10^{-5} , no seed detected, because a link that loses at the background rate is not a fault by the ledger's own definition. The flat portion of the curve is therefore a statement about the epoch volume relative to

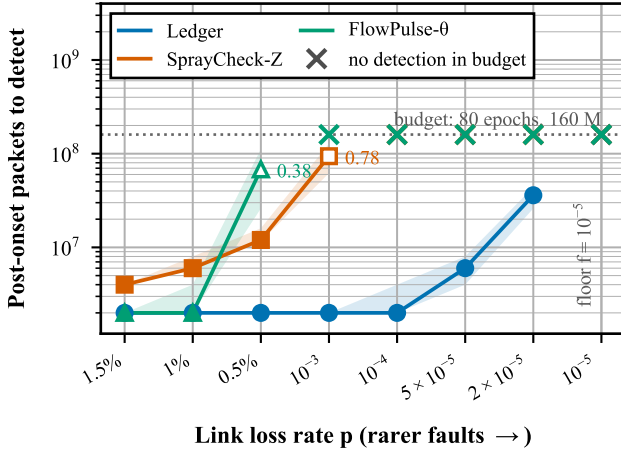


Fig. 2. Median post-onset packets to detect a single-spine grayhole against loss rate p , over 50 seeds, $k = 8$ spines, and a budget of 80 post-onset epochs of 2 million packets (dotted ceiling); shaded bands are interquartile ranges. Filled markers are cells with an action rate of 1.00; hollow markers are annotated with the fraction of seeds that detected, and their median is over those seeds only; a cross on the ceiling marks a cell in which no seed detected. The ledger's cost is one epoch until the loss rate nears the floor, then rises to its own wall; the passive detectors' cost grows toward the ceiling and crosses it two decades earlier.

TABLE I

DETECTION OF A SINGLE-SPINE GRAYHOLE WITHIN A BUDGET OF 80 POST-ONSET EPOCHS (160 MILLION PACKETS), 50 SEEDS PER CELL: ACTION RATE (ACT.) AND, OVER THE SEEDS THAT DETECTED, THE MEDIAN POST-ONSET PACKETS TO THE FIRST CORRECT FLAG (PKTS). THE 95 PERCENT WILSON INTERVAL IS [0.93, 1.00] FOR EVERY 1.00, [0.65, 0.87] FOR 0.78, [0.26, 0.52] FOR 0.38, AND [0.00, 0.07] FOR EVERY 0.00. FALSE POSITIVES WERE 0.00 FOR EVERY ARM AT EVERY RATE.

p	Ledger		SprayCheck-Z		FlowPulse- θ	
	Act.	Pkts	Act.	Pkts	Act.	Pkts
1.5%	1.00	2M	1.00	4M	1.00	2M
1.0%	1.00	2M	1.00	6M	1.00	2M
0.5%	1.00	2M	1.00	12M	0.38	68M
10^{-3}	1.00	2M	0.78	94M	0.00	—
10^{-4}	1.00	2M	0.00	—	0.00	—
5×10^{-5}	1.00	6M	0.00	—	0.00	—
2×10^{-5}	1.00	36M	0.00	—	0.00	—
$10^{-5} (= f)$	0.00	—	0.00	—	0.00	—

the floor, and the paper no longer calls it flat without that qualification.

The passive cost grows as $1/p^2$ and the action rate collapses. SprayCheck-Z needed two, three, and six epochs at 1.5, 1.0, and 0.5 percent. At 10^{-3} its median rose to 94 million packets, 47 times the ledger's, and that median is over the 78 percent of seeds that detected at all; its interquartile range widened from 4 to 4 million at 1.5 percent to 62 to 128 million at 10^{-3} . At 10^{-4} and below no seed detected within the budget. FlowPulse- θ failed one order of magnitude earlier, at 38 percent of seeds and 68 million packets at 0.5 percent and nothing below. Both passive arms had a false-positive rate of 0.00 at every rate, so the collapse is a loss of sensitivity, not a trade against false alarms. The 22 percent SprayCheck missed at 10^{-3} and its whole column at 10^{-4} are the same s^2/p^2 wall pushed past 160 million packets. A larger budget

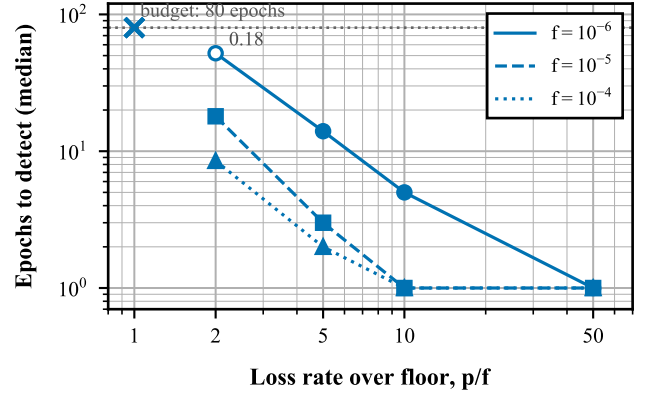


Fig. 3. The ledger's wall tracks the floor: median epochs to detect against the loss rate as a multiple of the background floor f , for three floors, 50 seeds per cell. A cross at the top marks a cell in which no seed detected within 80 epochs; a hollow marker is annotated with its action rate. At every floor the ledger detects nothing at $p = f$, detects in every seed from about $5f$ upward, and reaches one epoch by $10f$ to $50f$.

would move the passive floor to a rarer p but cannot change the exponent. This is a statement about cost, not capability: a passive detector with an unbounded budget can eventually detect at 10^{-4} , and nothing here says otherwise.

The wall moves with the floor. Figure 3 repeats the ledger arm at three background floors, 10^{-6} , 10^{-5} , and 10^{-4} , over p from f to $50f$. The curves align when plotted against p/f . At $p = f$ nothing is detected at any floor. At $2f$ the action rate is 1.00 at the two higher floors, in a median of 18 and 8.5 epochs after onset, and 0.18 at 10^{-6} , where the floor produces half a drop per epoch and the separation is the sparse Poisson case of the proposition. From $5f$ upward every seed detects, in a median of 14, 3, and 2 epochs after onset at floors of 10^{-6} , 10^{-5} , and 10^{-4} , falling to one epoch by $50f$, $10f$, and $10f$. Consequently, we expect the loss rate at which the witness stops paying to be not a fixed number but a small multiple of whatever the fabric's background floor is, which production floors can put anywhere from 10^{-6} to 10^{-4} [5]. SprayCheck's action rate at a fixed p barely changes with the floor, since its noise is spray noise rather than floor noise: 0.16 at $p = 5 \times 10^{-4}$ over a 10^{-5} floor and 0.06 over 10^{-4} .

What the two bytes buy: in-band alignment. Figure 4 answers the question a per-link counter pair raises. At $\sigma = 0$ the counter pair reproduced the ledger cell for cell, 2 million packets and no false positive at every rate, including the ledger's own wall at $p = f$, where both arms detect nothing; that column confirms that Equation (3) contains no information a counter pair lacks. The two arms separate as soon as the two registers are read at different instants. At $\sigma = 10^{-4}$, ten microseconds against a 100 ms epoch, the counter pair still flagged the faulty spine in every seed. It also flagged a healthy spine in 28 percent of seeds at 0.5 percent loss and above, in 94 percent at 10^{-3} , and in every seed at 10^{-4} . At $\sigma = 10^{-3}$ and above, including the measured best case of $\sigma = 0.026$, it flagged a healthy spine in every seed at every rate. At the floor, where there is no fault, it still raised an alarm in 62 percent of seeds at $\sigma = 10^{-4}$ and in every seed above that. The reason is

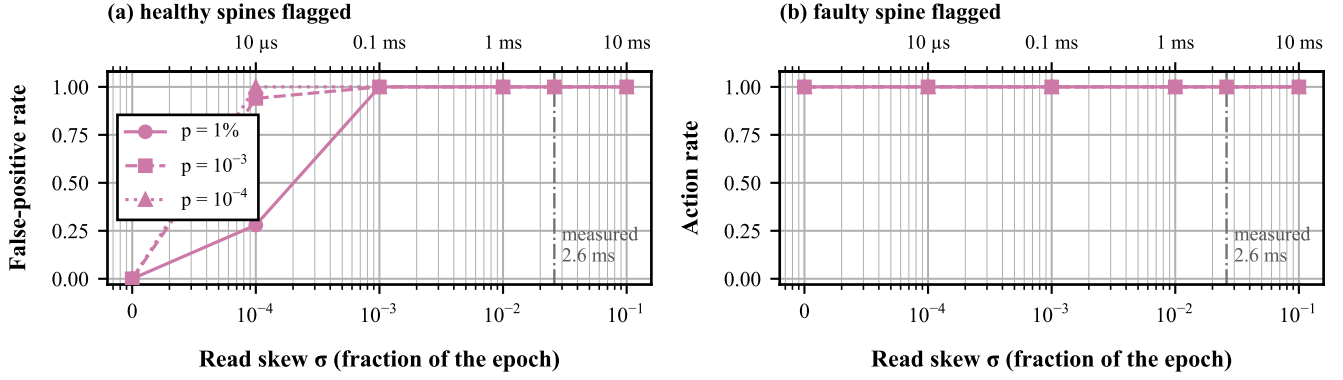


Fig. 4. The zero-byte counter pair against controller read skew, 50 seeds per cell, for three loss rates; the upper axis gives the skew in milliseconds for the harness’s 100 ms epoch. (a) Fraction of seeds in which a healthy spine was flagged. (b) Fraction of seeds in which the faulty spine was flagged. At $\sigma = 0$ the arm is the ledger’s information read at one instant and matches the ledger exactly; the vertical line is the best case measured on the switch, a 2.6 ms pairwise read. For this arm the action rate must be read beside the false-positive rate: a cell at 1.00 in both panels is a failure in which nearly every link is named, not a detection.

that a read skew of σ on a link carrying λ packets per epoch adds phantom loss with standard deviation near $\lambda\sigma/\sqrt{6}$. That is about 10 packets at $\sigma = 10^{-4}$ and 2700 at $\sigma = 0.026$, against a true signal of 25 packets at 10^{-4} and a floor of 2.5. The arm does not lose the faulty link; it loses the ability to say which links are healthy, which we believe is the failure a repair action can least afford. Section VI shows the consequence for localization. The witness reads both registers on one switch in one batched operation, so its skew is zero by construction. That, and not any inference, is what two bytes per packet buy. A counter pair would need to match it by reading its two registers within about ten microseconds of each other across two switches, four orders of magnitude tighter than the 350 ms census and two orders tighter than the 2.6 ms pairwise read that we measured.

Two disclosures. SprayCheck’s sensitivity is calibrated under its paper’s Poisson model at 2.5 million packets per spine, the cumulative count its Z-test reaches after ten epochs; its real join-the-shortest-queue spraying has less noise, which would lower its curve by a constant factor without changing the exponent. The conditional medians of the hollow points in Figure 2 are survivorship-biased, which is why we read the action-rate collapse, not the slope of a censored median, as the load-bearing signal.

VI. LOCALIZATION (O2)

Detection answers whether something is wrong; localization answers which directed link, which is the question a repair needs. This section is a question about the passive detectors, not about the ledger. Given per-directed-link counts, a detector that flags a link has already localized it. The ledger’s row is therefore the oracle against which we measure three things: how far down in p a receive-only vantage can still disalias the uplink and downlink pair, what it costs, and what a counter pair read with skew does to the same information. Figure 5 presents the exact-localization rate and the size of the named set for a downlink fault and for an uplink fault, over 50 seeds per cell on the two-hop fabric of Section IV; every rate in this

section carries a 95 percent Wilson interval of [0.93, 1.00] at 1.00 and [0.00, 0.07] at 0.00, and the full table is in the artifact.

The oracle row, and the counter pair beside it. In every cell of both families the ledger’s exact rate was 1.00 [0.93, 1.00], its mean set size was exactly 1.00, and it named a healthy link in no seed. This is definitional: it detects in every seed with a false-positive rate of 0.00 (Section V), and the sublink it rejects is a directed link. The counter pair read at one instant reproduced the row exactly. The counter pair read with the measured best-case skew of $\sigma = 0.026$ named a set of about three links in every seed and never the faulty link alone. The reason is that the phantom loss of Section V makes two or three healthy links look faulty beside the real one; the same information, read on two switches, no longer localizes. The measured result of this section is therefore the passive arms’ aliasing and the counter pair’s skew, with the ledger as the reference.

SprayCheck-Z ties at high loss and degrades to the pair at low loss. At 1.5 percent in both families and at 1.0 percent for the uplink, SprayCheck’s cross-leaf intersection resolved the fault to one directed link as well as the ledger did. There were no discordant seeds between the two arms in those cells, and we report the tie as a null result rather than as a test statistic, since a McNemar test with no discordant pairs is uninformative. At 0.5 percent its exact rate fell to 0.34 for the downlink and 0.26 for the uplink. At 10^{-3} it fell to 0.00 and 0.02 with a mean set size of 2.00, i.e., the uplink and downlink pair its paper says an uncorroborated report is left with. The reason is that at those rates its Z-test flags the faulty spine on too few distinct leaves to complete the intersection, so it can name the spine but not the directed link. Every such pair contains one healthy link, which is why the ambiguous column reaches 1.00 there: a repair that acted on the set would touch the wrong hop half the time. At 10^{-4} it missed in 74 percent of seeds even under the generous budget.

FlowPulse- θ cannot localize an uplink fault at any rate. For a downlink fault it was exact in 94 percent of seeds at 1.5

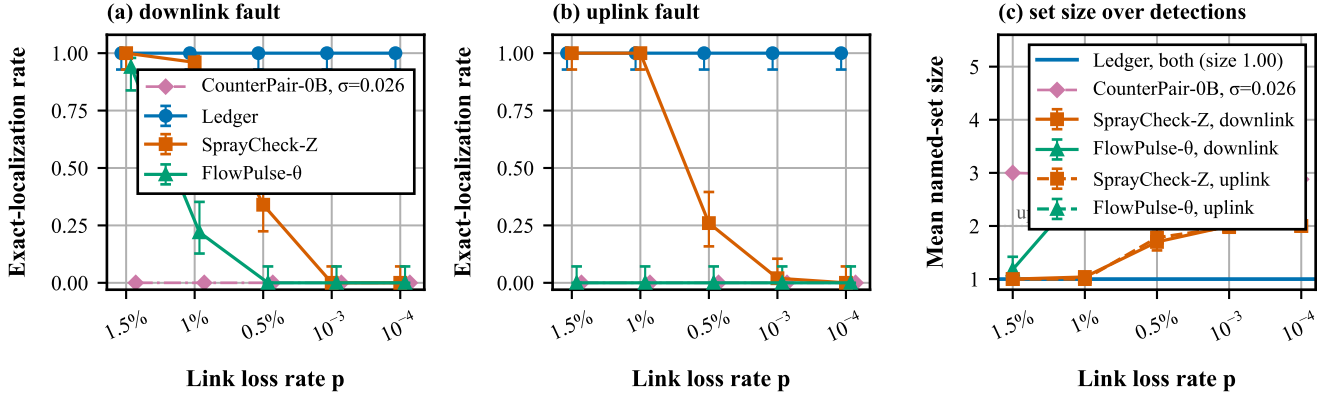


Fig. 5. Localization under spraying, 50 seeds per cell with 95 percent intervals. (a) Exact-localization rate for a downlink fault. (b) Exact-localization rate for an uplink fault. (c) Mean size of the named set over detections: 1 is the exact directed link, 2 is the uplink and downlink pair; the ledger's set size is 1.00 in every cell, and a missing point means the arm detected nothing.

percent, fell to 22 percent at 1.0 percent with a mean set size of 2.74, because its all-or-one rule landed in neither branch and returned the port's whole candidate set, and missed at 0.5 percent and below. For an uplink fault it missed in every seed at every rate, including 1.5 percent. The reason is that a single sender's loss moves the aggregate load of the ingress port by only about $1/(\text{number of senders})$ of the drop rate, here about 0.5 percent for a 1.5 percent uplink loss shared by three senders, which never clears the paper's fixed one-percent threshold. This is faithful to FlowPulse's own design: in its native setting one sender occupies each port and an uplink fault is the whole port's load. Under multi-sender spraying the same fault is diluted below its threshold. The gap is a property of the sprayed fabric, reproduced, not a handicap imposed on the baseline.

VII. THE LEDGER ON SILICON (O3)

The replay harness models the ledger's counters. This section measures them. Because the passive detectors are host-side systems that cannot run on our fabric, silicon cannot host the head-to-head; what it can do is show that the ledger's own side of the comparison is measured rather than modelled, at the same loss rates, on a real switch. We expect the ledger to recover the injected loss exactly at every rate, on the first read after the traffic, and to attribute it to the injected directed link alone; a one-packet discrepancy may appear where a loss trails the last survivor of a read.

Recovery across the loss regime. Table II and Figure 6(a) present nine cells on one uplink sublink. In every cell the recovered loss equals the armed drop count exactly, from 50 drops in 5000 packets down to 6 drops in 60000, and the two clean cells, together more than 42000 packets, recovered 0. The ledger reported each value on the first census read after the traffic. It does not accumulate packets toward a statistical threshold, because Equation (3) counts the missing packets rather than inferring them. This is the structural contrast with Section V. At 10^{-4} , where the passive detectors detect nothing within the budget, the witness recovers the six lost packets on real hardware.

TABLE II
RECOVERY OF INJECTED LOSS ON THE TOFINO SWITCH, ONE SUBLINK, EXACT INJECTOR: PACKETS SENT N , DROPS ARMED D , REGISTER DELTAS, AND THE RECOVERED LOSS $\Delta_{seq} - \Delta_{obs}$. EVERY CELL RECOVERED D EXACTLY, AND THE TWO CLEAN CELLS RECOVERED 0.

D/N	N	D	Δ_{seq}	Δ_{obs}	Recovered	Match
clean	20000	0	20000	20000	0	yes
clean	20000	0	22050	22050	0	yes
10^{-2}	5000	50	5000	4950	50	yes
10^{-2}	5000	50	5000	4950	50	yes
10^{-3}	30000	30	30000	29970	30	yes
10^{-3}	30000	30	30000	29970	30	yes
5×10^{-4}	40000	20	40000	39980	20	yes
10^{-4}	60000	6	60000	59994	6	yes
10^{-4}	60000	6	60000	59994	6	yes

Random drops. A scheduled burst is the easy case, so we repeated the measurement with the stochastic injector, which drops each packet independently with probability $W/2^{16}$, and we compared the recovered loss with the injector's own counter of packets it dropped. At a target of 10^{-4} ($W = 7$, realized rate 1.07×10^{-4}) on 60000 packets, the ledger recovered 7 and the injector counted 7. At a target of 10^{-3} ($W = 66$, realized 1.01×10^{-3}) the ledger recovered 66 and the injector counted 67. Two mechanisms, each a testable claim, account for the one-packet gap; the run that would discriminate between them, a repeat with a clean tail burst that guarantees a trailing survivor before the read, was not available on the shared switch, so we state both rather than assert either. First, a drop with no later survivor on the sublink is not exposed until the next survivor advances the frontier, which is the trailing-loss property of Section III. Second, on that uplink egress the injector's offered count ran about 180 to 190 packets above the 60000 host departures. In other words, a small population of packets that are not part of the host-stamped sequence crossed the same egress, and the injector saw them but the ledger did not. The recovered value tracks the loss of stamped packets, which is the quantity the ledger is defined to recover.

Localization on the uplink. Figure 6(b) shows both at-

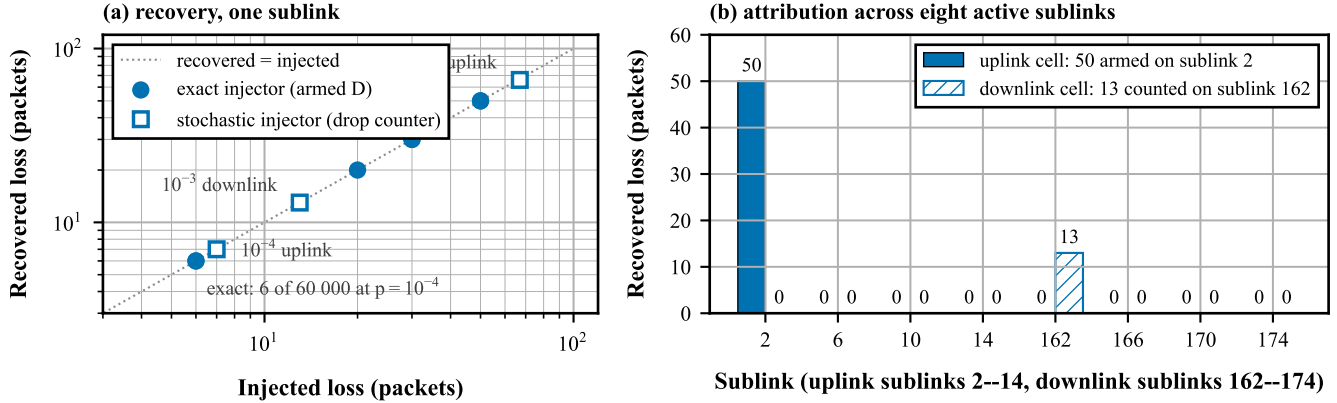


Fig. 6. The ledger on the Tofino switch. (a) Recovered loss against injected loss on one sublink, on logarithmic axes, for the exact injector, whose ground truth is the armed count, and for the stochastic injector, whose ground truth is its own drop counter; every point lies on the identity except the 10^{-3} uplink cell, one packet below it. (b) Recovered loss on each of the eight active sublinks for the uplink cell and for the downlink cell; only the injected sublink recovers a non-zero loss in each.

tribution cells. We armed 50 drops on one uplink sublink and sent 20 000 packets across all four traffic classes, 5 000 each, so that every one of the eight tracked sublinks carried traffic. The injected sublink recovered 50, and the other seven recovered 0. Among the seven is the downlink sublink of the same traffic class, which showed 4 950 arrivals rather than 5 000, because the 50 packets dropped on the uplink never reached the downlink to be stamped. It manufactured no loss there, since its Δ_{seq} and Δ_{obs} advanced together.

Localization on the downlink. The mirror case is the one the passive vantage cannot resolve. We armed the stochastic injector at 10^{-3} on a downlink sublink and sent 80 000 packets across all four classes, 20 000 each. The downlink sublink recovered 13 and the injector counted 13. Every other sublink recovered 0, including the uplink sublink of the same class, whose 20 000 packets had all been witnessed before the downlink drop. On silicon, then, a downlink grayhole is attributed to the downlink directed link alone and an uplink grayhole to the uplink alone, with zero false attribution on seven active sublinks in each case. This is the disaliasing property the replay harness measured in Section VI.

Scope. These cells measure recovery fidelity and false attribution for one independent fault at a time, and three things about them should be read plainly. The fabric is virtual: four leaves and two spines emulated on one Tofino with a four-lane loopback cable, each directed link one traffic-manager queue. There is no second physical switch, no physical spine, and no cross-traffic, so the cells validate the register arithmetic and the link reconstruction on real silicon, not a multi-switch deployment. The offered load was 20 thousand packets per second of 1400-byte packets, about one percent of a 25 Gb/s port. The register update rate at line rate is asserted from the pipeline placement, not measured; a line-rate soak is future work, and until it is done the silicon claims are stated at the measured rate. The recovered loss is magnitude-based and is the same for a burst and for dispersed drops of the same count; the one-gap-per-discontinuity property is covered by the model tests of Section IV, not here. The added-load figures of

Section IX are computed from the header size, not measured as saturating throughput.

VIII. THE BOUNDARY: CORRELATED FAULTS (O4)

Every result so far injects one independent fault. This section asks whether the advantage survives several faults at once, whether it survives a correlated, fleet-wide shift in loss, and whether it survives persistent congestion loss on a subset of links. The last is the shape of incast at a collective's reduction point, and it is more likely in a training fabric than a uniform shift. It is a gate rather than a pass: where the advantage collapses, that bounds the headline claims, and the first two regimes were pre-registered as that boundary. The harness of Section IV drives four regimes over the 64 directed links of the two-hop fabric, 50 seeds per cell and 40 post-onset epochs, with the ledger's controller unmodified. Two anchors accompany every cell: a do-nothing arm that never acts, and an oracle that returns the true set. The injected loss was verified in the data of every cell, not in the configuration. Figure 7 summarizes the four regimes; the full per-cell tables, with intervals and both the window and final-epoch scores, are in the artifact.

A. Several Independent Faults

Figure 7(a) presents the first regime, in which $M \in \{2, 3\}$ distinct directed links are faulty at the same rate against a healthy background. The ledger named every one of the faulty links on the first epoch after onset, with no false link, at both 0.5 percent and 10^{-3} . The reason is that it judges each directed link on that link's own evidence, so a second fault elsewhere in the fabric changes nothing about the first. SprayCheck-Z became worse as faults multiplied. Because several simultaneous failure reports create spurious cross-leaf intersections, it recalled the true faults at 0.5 percent but added 1.2 to 1.4 healthy links per seed. It named exactly the true set in only 18 to 28 percent of seeds. At 10^{-3} its recall itself fell to 0.63 to 0.75, and a healthy link was still named at

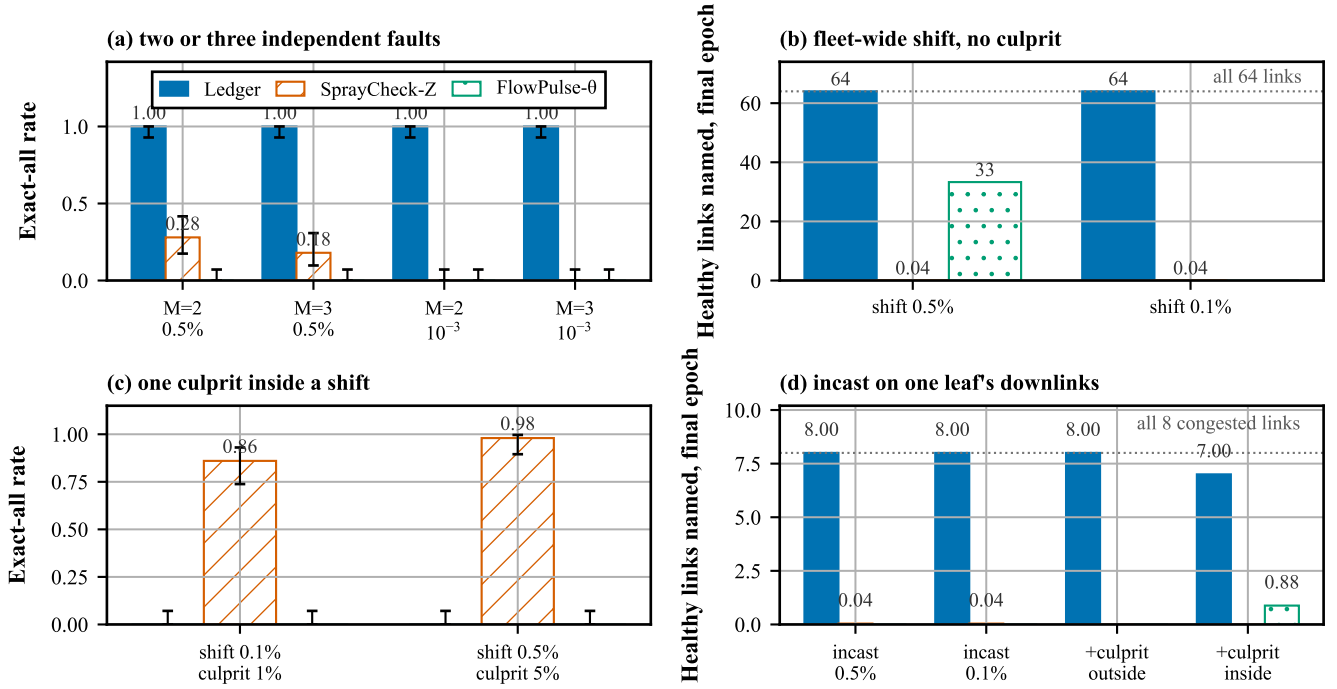


Fig. 7. The correlated-fault gate, 50 seeds per cell over 40 post-onset epochs on 64 directed links. (a) Exact-all rate, with 95 percent intervals, for two and three simultaneous independent faults. (b) Healthy links falsely named at the final epoch under a fleet-wide shift with no culprit, where the correct answer is zero and 64 is every link. (c) Exact-all rate for one culprit embedded in a shift. (d) Healthy links falsely named at the final epoch when the eight downlinks into one leaf carry persistent congestion loss, alone and with a culprit outside or inside the congested set. The ledger holds in (a) and fails in (b), (c), and (d), where SprayCheck-Z holds.

the final epoch in about half the seeds. FlowPulse- θ missed at these rates, as in the single-fault sweep. The independent-fault advantage therefore appears stronger with several faults than with one, and the spurious-intersection behaviour is a limitation of the passive localizer that we have not found reported elsewhere.

B. A Fleet-Wide Shift With No Culprit

Figure 7(b) presents the second regime, in which every directed link's loss rises to the same level and no link is individually at fault, so the correct answer is to name nothing. The ledger failed this gate. It named all 64 directed links in every seed, on every epoch through the end of the window, which is worse than doing nothing. The reason is that its fleet floor is a leave-one-out estimate that lags a shift shared by every link. Each link's ratio then reads the shifted but healthy rate as strong evidence, and e-BH rejects nearly every link. The rejected links leave the pool from which the floor is estimated, and the fleet latches. We had predicted this failure from our own analysis of the decision rule before the gate was run, and the gate reproduced it on fresh seeds. SprayCheck-Z was the arm that held here, with a false-positive rate of 0.02 at the final epoch. Its Z-test normalizes each flow by that flow's own observed count, which falls with the shift, so a uniform shift can cancel. FlowPulse- θ flagged nearly every port when the shift's two-hop deficit crossed its one-percent threshold and flagged nothing when it did not; below the threshold it is blind rather than resistant.

C. A Culprit Inside the Shift

Figure 7(c) presents the third regime, a shift plus one link worse than the shifted background. The ledger did find the culprit, in every seed and on the first epoch, but it drowned the culprit among the other 63 links, so its exact-all rate was 0.00 and its answer was not actionable. SprayCheck-Z isolated the culprit cleanly against the shifted background, with exact-all rates of 0.86 and 0.98 and no healthy link named at the final epoch, for the same reason that let it hold in the second regime. In this regime, then, the exact directed-link localization that is the ledger's advantage under independent faults becomes SprayCheck's advantage.

D. Congestion Loss on a Subset of Links

Figure 7(d) presents the fourth regime. The eight downlinks into one leaf carry a persistent elevated loss of 0.5 or 0.1 percent while the other 56 links stay at the floor, first with no fault anywhere and then with a 1 percent culprit on an uplink outside the congested set or on a downlink inside it. This is congestion loss, not a fault, and the correct answer names none of the eight congested links. The ledger named all eight, in every seed, at every epoch, and with a culprit present it found the culprit in every seed but could not separate it from the seven or eight congested links beside it, so its exact-all rate was 0.00. The reason is not the floor this time, which the 56 healthy links keep well estimated; it is that the ledger is a loss counter, and congestion loss is loss. Equation (3) counts the packets a queue dropped exactly as it counts the packets a

TABLE III

COST OF THE LEDGER BESIDE THE SELF-REPORTED COST OF THE PASSIVE DETECTORS. LEDGER NUMBERS ARE FROM THE HARDWARE COMPILER ON SDE 9.13.2, WHICH MATCHED THE 9.13.1 COMPILER EXACTLY; THE 4-BYTE COLUMN IS THE EARLIER VERSION THAT ALSO CARRIED A LINK IDENTIFIER. SPRAYCHECK'S STATE IS UNDER 2 KB IN TOTAL BUT COVERS ONE FLOW AT A TIME, ROTATING ABOUT ONCE A MINUTE; FLOWPULSE'S PAPER STATES NO MEMORY FIGURE. WE HAVE NOT REIMPLEMENTED EITHER IN P4, SO NO STAGE COUNT EXISTS FOR THEM.

	Ledger, 2 B	Ledger, 4 B	Passive
Wire bytes per packet	2	4	0
Added load (1400 B)	0.14%	0.28%	0
State per directed sublink	6 B	6 B	see caption
Sublinks covered at once	1024	1024	one flow / port
Ingress / egress stages	12 / 5	11 / 5	—
SRAM / TCAM blocks	91 / 16	89 / 15	—

damaged optic dropped, and nothing in the decision rule asks why. SprayCheck-Z held again, naming a congested link at the final epoch in 2 percent of seeds and isolating the culprit in 92 to 96 percent, because its per-flow normalization cancels a loss that every flow through the congested leaf shares. FlowPulse- θ was blind to the congestion at both levels, as its per-port deficit stayed below its threshold, and named three links when the culprit sat inside the congested set. We had expected this result from the analysis of Section III, and we report it because an operator is likely to meet incast far more often than a uniform fleet-wide shift. Section X states what a fix needs: a signal that separates queue loss from link loss, which the testbed's queue-depth metadata can supply and which this paper does not evaluate.

E. What the Gate Decides

The advantage holds, and widens, for several simultaneous independent faults; it fails under a common-mode shift, and it fails under persistent congestion loss on a subset of links. Consequently, the claims of Sections V to VII are stated for one or several independent per-directed-link faults on a fabric whose remaining links lose packets only at the background floor. We state the converse as plainly. Where loss is shared, whether by every link or by the links behind one congested queue, the ledger's absolute test names the shared loss as faults, and a relative passive test is the better tool. Section X discusses what a fix would need.

IX. COST

The passive detectors pay nothing on the wire, and the ledger does. This section reports the ledger's cost in the currency each baseline paper uses for its own, from the compiler's resource report on the switch's own software development kit, not from an estimate. Table III presents it beside the numbers the two baseline papers report for themselves.

Wire. The witness is two bytes per packet. Against the bytes a packet occupies on the wire, including Ethernet framing, preamble, and inter-frame gap, that is 0.14 percent for a 1400-byte payload and 0.02 percent for a 9000-byte jumbo frame, but 0.7 percent for a 256-byte packet and 2.4 percent for a 64-byte one. In a fabric where a non-trivial share of packets are

small acknowledgements, the honest figure is that range, not the 1400-byte point. The earlier four-byte version, which also carried a link identifier, cost twice each figure, the same order at 1400 bytes as the ± 0.25 percent flow-priority distortion SprayCheck reports for its own prioritized flows [7]. The stamp is stripped before host delivery, so the cost is bandwidth on every fabric hop and nothing at the endpoints.

Sequence width and read cadence. The 16-bit sequence wraps every 65 536 packets per sublink, so Equation (3) is unambiguous only if the controller reads a sublink at least once per wrap. We measured the controller's read loop on the switch: one sublink's two registers take 2.6 ms to read, and a census of all 1024 sublinks takes 350 ms (median of five, range 304 to 358 ms), timed on the switch's own host and so excluding an off-box controller's network round trip. At 25 Gb/s with 1400-byte payloads a port carries about 2.2 million packets per second, and a sublink carrying a whole port wraps in 30 ms. Consequently, with a 350 ms census the two-byte witness can be unambiguous only while a sublink carries under about 190 thousand packets per second, roughly 9 percent of a port at that packet size. The silicon cells of Section VII ran at 20 thousand packets per second, 3.3 s per wrap, and are unaffected. Above that rate the read loop must be shortened, for example by reading hot sublinks individually, or the sequence widened to 32 bits, which is four bytes of sequence rather than the earlier four-byte header, whose extra bytes were a link identifier and not width. We note that the earlier header therefore does not escape this bound either; both widths wrap at 2^{16} . The same 350 ms census therefore limits both arms of Section V, for different reasons. It makes the counter pair fabricate loss that nothing distinguishes from real loss, and it caps the witness at about 9 percent of a port; but the witness's failure is a bounded modular-arithmetic condition the controller can see coming and avoid by reading a hot sublink individually at 2.6 ms, while the counter pair's is not.

State. Each directed sublink holds a 16-bit sequence register and a 32-bit observed-count register, six bytes, and the program tracks all 1024 sublinks continuously. SprayCheck's own state is under two kilobytes, but it instruments one flow at a time and takes about a minute to rotate to the next [7]; FlowPulse's paper does not state a memory figure [8]. The trade is continuous per-link coverage of the whole fabric for a fixed per-packet tag, against zero tag for a rotating sample.

Pipeline. Halving the wire cost was not free. Reconstructing the directed link at ingress from the port and the spray field added two narrow tables, and the compiler placed them in a new stage. The program went from 11 to 12 ingress stages, which is the hardware ceiling of the Tofino 1, with 5 egress stages unchanged. It also went from 89 to 91 SRAM blocks and from 15 to 16 TCAM blocks. The compile was byte-for-byte identical on the laptop SDE 9.13.1 and the switch's 9.13.2, with zero errors and the same five warnings on both. The stage cost follows from a choice we made before compiling. To avoid destabilizing logic that other parts of the program rely on, we added narrow tables rather than widen the key of an existing multi-purpose port-role table. The program now has no ingress stage headroom, and any future ingress addition must either remove a stage elsewhere or revisit that

choice by folding the reconstruction key into the port-role table. We report this as the real cost of the two-byte witness rather than as a footnote, because a reviewer who weights pipeline stages over wire bytes can reasonably prefer the four-byte version.

X. DISCUSSION AND LIMITATIONS

What the detection buys at the application. The paper's motivation is that a grayhole at 10^{-3} to 10^{-4} slows every collective that crosses it, and the reader is owed the measurement that tests it. We ran one, and it is unfavourable to the motivation as first stated. We simulated a mixture-of-experts collective at packet level on a 1024-uplink sprayed fabric, with a 10^{-3} grayhole injected on a fully loaded uplink, 255 silent drops over the job, and retransmission-timeout recovery at its worst-case setting of 40 ms. The collective's completion time rose by 1.36 percent over the fault-free run (3.588 s to 3.636 s). Because the timeout was at its largest value, the figure should bound every faster recovery setting. Because the simulator's measurement layer has no mitigation actuator, the fault-free run is the only ceiling on what any mitigation could recover: at most 1.36 percent at this operating point. About 99.5 percent of the drops appear to have been absorbed by the collective's pipelining, since a timeout on one chunk overlaps other chunks' slack and only the worst straggler moves the barrier. Consequently, the value of naming a 10^{-3} link is not the completion time of the collective that is running when it is named. It lies elsewhere, and we state where without claiming more than the measurement supports. First, a link losing at 10^{-3} from a physical cause is a link that is degrading toward 10^{-2} or toward a blackhole; Section II already assumes, with CorrOpt [5], that such causes persist until repaired, so early naming has repair and attribution value that a single completion-time measurement cannot capture. Second, the 1.36 percent is on one collective, and a fleet runs thousands of links continuously, so the aggregate is not 1.36 percent of one job. Last, the contribution of this paper is the measurement instrument and its boundary. The application-level value of low-loss detection is an open question, and our own simulation bounds it at about 1.4 percent per collective under timeout-dominated recovery.

Common-mode and congestion loss. The failures of Section VIII are the ledger's largest limitation, and they are properties of the decision rule, not of the witness. The registers still recover each link's loss exactly under a fleet-wide shift and under incast. What fails is a comparison against a floor that moves with the fault in the first case, and the absence of any question about the cause of the loss in the second. A fix for the first may need a guard that detects when the floor itself is drifting, or a corroboration gate that requires a link's ratio to stand out relative to its neighbours as well as relative to the floor, i.e., the relative normalization that lets SprayCheck hold in that regime. A fix for the second needs a signal that separates queue loss from link loss; the testbed exposes per-packet queue depth in the egress pipeline, so a stamp that carries it, or a gate that consults it, is the obvious candidate. Both are design work with their own evaluation, and this paper does not do them.

What the two bytes buy, and what they do not. Equation (3) contains no information a zero-byte counter pair lacks, and Section V shows the two are identical when the pair is read at one instant. What the stamp buys is that the sender's count arrives in band, so the receiver holds both terms of the difference and reads them in one operation. The pair's tolerance to read timing is about ten microseconds across two switches, against a measured 2.6 ms for the nearest pairwise read and 350 ms for a census on our controller. The stamp does not buy a better inference: the localization score of 1.00 in Section VI is definitional given per-directed-link counts, and we present the ledger there as the oracle row against which the passive detectors' aliasing is measured, not as a result.

A synchronized counter snapshot. The read-skew argument invites an obvious reply: do not read the two registers at one instant, latch them at one instant and read them at leisure. If both switches copied their per-directed-link counters into shadow registers on a common epoch boundary derived from precision time, the controller could take its full 350 ms to collect them. The skew term would then be sub-microsecond, an order below the tolerance Section V establishes, and the counter pair would match the witness at zero wire bytes. This is the alternate-marking family of loss measurement [15], and RFC 6374 is built around it. We believe the reply holds in principle, and we scope the paper against it. Three things bound it in practice. First, the snapshot is not free in the data plane, and our own history prices it. The program this ledger replaced kept alternating counter banks per epoch with a guard interval. The compile gate that retired it measured the banked scheme at two more tables, three more SRAM blocks, one more stateful ALU, and two more header containers than the ledger at the same stage count, before any time synchronization. It also carried the bank parity in a fabric header byte, so it was not zero-byte either, and a transit stage that failed to preserve that byte produced a 56 percent false-blackhole rate until it was found. A snapshot that keys on time rather than on a carried bit avoids the byte but not the banks. Second, the witness needs no fabric-wide coordination: no time distribution, no trusted common time base, and no agreement between two switches about when an epoch began. That is the operational asymmetry that survives the reply. Third, and stated plainly, an operator who has precision time on every switch and can afford the second register bank and the bank swap should prefer the synchronized counter pair, and the witness is then unnecessary. We did not build the synchronized variant; its stage cost on this program is an estimate from the banked scheme we did build.

The wire reduction, reconsidered. Halving the header from four bytes to two cost the program's last ingress stage (Section IX), and the soak below shows it also introduced an anomaly the four-byte program did not have and a structural aliasing of traffic that carries no real spray choice. Neither header escapes the 16-bit wrap bound, since both carry a 16-bit sequence. An operator who weights pipeline headroom or the open anomaly over one byte per packet per hop can reasonably prefer the four-byte program, and the evidence in this paper now points that way rather than merely permitting it. We report both because the trade should be made rather than assumed.

The anomaly on unmeasured sublinks, as far as the evidence goes. During a 57-cycle soak of the two-byte program, the injected-loss recovery passed in every cycle. However, two sublinks that were neither armed nor measured each showed one stamped packet with no matching arrival during clean traffic, a signature not seen across about 3 200 historical cycles of the four-byte program. We took the investigation as far as the shared hardware allowed. Physical-layer loss was ruled out by port-level media-access counters (9 224 frames sent and received) and ambient noise by a fully idle window that moved no register. A byte-identical build of the four-byte source, deployed through the same pipeline and run with the same script immediately afterward, showed 0 anomalies in 100 cycles against 2 in 57 for the two-byte build, which isolates the anomaly to the wire-reduction build rather than the testbed. A probe with no injector table writes reproduced it, which rules out the control-plane writes that accompanied the first two occurrences. An idle-then-burst trigger looked confirmed at 3 of 5 post-idle bursts, then two further runs with the corrected instrument found 0 of 9. The honest count is therefore 3 of 14, and the surviving hypothesis is a genuinely cold fabric after a fresh program load rather than mid-session idleness. The root cause is not established, and the fresh-load protocol that would settle it needs its own bring-up budget. Every number in this paper comes from armed, measured sublinks with baseline and post-traffic reads, and none is affected. However, a claim of zero false positives must be read beside a one-packet phantom loss that recurs on the same program, at a rate near 2 per 57 cycles, on unmeasured sublinks.

A structural aliasing introduced by the wire reduction. While restoring the two-byte program after the comparison above, a census of traffic classes outside the probe's set showed phantom arrivals with no stamps on two classes, reproduced across two independent bring-ups. The explanation is mechanistic. Bring-up's own port-verification frames carry no real spray choice and arrive with a spray field of zero, and the new ingress reconstruction maps every frame with spray zero to the first virtual link regardless of which leaf sent it, where the old design read the link identifier the sending egress had written. On the testbed this never touched a measured sublink. In a production fabric, control and management traffic that is stamped but not sprayed would be attributed to one link's counters, and the two-byte design needs either a table entry that excludes unsprayed traffic from the stamp or a spray value reserved for it. We record this as a correctness cost of the reduction that Section IX did not price.

Silicon scope. Section VII states the limits of the silicon cells: a virtual fabric, one percent of a port, one sublink, computed rather than measured added load, and a one-packet discrepancy explained but not re-measured. The stage budget is that of a stripped program, forwarding plus the ledger. A production leaf program that also carries load balancing, transport offload, access control, tunnelling, and quality of service does not have a spare stage on a Tofino 1, and whether the ledger fits beside such a program, or on a Tofino 2, is not measured here.

Generality. The replay fabric has four leaves and eight spines with uniform spray. FlowPulse's dilution grows with

senders per port and SprayCheck's intersection improves with leaves, so we expect the ordering of the arms to survive a change of radix while the crossover rates are properties of this fabric. The $1/p^2$ form of Proposition 1 does not depend on the topology; its constants do, through the floor and the epoch volume, and Section V sweeps both.

XI. RELATED WORK

Table IV positions the ledger against the systems closest to it on the axes this paper measures. We group them into four families.

In-fabric loss witnesses. NetSeer [9] inserts a per-neighbour sequence number at egress and treats an inconsecutive number at the downstream ingress as a drop, which is exactly the ledger's encoding at four bytes. LinkGuardian [10] adds an era bit and a dummy packet and retransmits the lost packet across the link. LossRadar [11] stores packet digests in invertible Bloom filters at both ends of a link and decodes the difference, which its authors state works per packet only below about 0.1 percent loss. The Ultra Ethernet link-layer retry [2] places the sequence in the MAC, and RFC 6374 [16] pairs transmit and receive counters per measurement entity. These systems own the primitive this paper uses, and we claim none of it. The counter pair of RFC 6374 is, in the information it holds, the same as the witness; Section V measures the read skew that separates them, which Table IV calls skew-limited and which no prior system quantifies. Compared to this work, none of them reports what the witness buys or costs on a sprayed fabric at 10^{-3} to 10^{-4} , measured against the passive detectors built for that fabric.

Passive detectors for sprayed fabrics. SprayCheck [7] and FlowPulse [8] are the direct comparators of this paper and the state of the art for sprayed fabrics. Both are zero-byte, both are deployable on existing counters, and SprayCheck reports zero added network load. The ledger does not dominate them on added load, because it pays two bytes per packet and they pay nothing. What it buys, measured in Sections V and VI, is a detection cost that does not grow as $1/p$ and a localization that does not alias the uplink and downlink pair, and both follow from holding each directed link's own transmit count. Under a common-mode shift, SprayCheck's relative test can hold where the ledger's absolute one cannot (Section VIII). OmniPath Ping [20] is concurrent work that addresses path ambiguity under spraying with an in-network per-hop cache and active probes; it is a tracing tool rather than a cost study, and it owns the problem statement of active measurement under spraying.

Active probing and tomography. dShark [17] localizes drops, including silent ones, by analysing mirrored per-hop traces on hosts after the fact, at a cost of mirroring the traffic and of capture drops that can hide real drops. The ledger's only edge over it is that two registers per sublink on a rerouteable resource are cheaper and act in place, not that it can find anything dShark cannot. NetBouncer [18], 007 [21], deTector [22], Pingmesh [23], and Everflow [24] probe or trace paths from hosts; under per-packet spraying a probe cannot pin its path, so they localize to a path or a scalar per link

TABLE IV

PRIOR SYSTEMS ON THE AXES THIS PAPER MEASURES. DL: DIRECTED LINK. “NOT ADDRESSED” MEANS THE PAPER DOES NOT TARGET THAT AXIS. THE LEDGER ROW RESTATES MEASUREMENTS FROM SECTIONS V TO IX; EVERY OTHER CELL IS TAKEN FROM THE CITED PAPER.

System	Primitive	Vantage	Per-DL	Under spraying	Low-loss detection cost	Localization under spraying	Restoration
NetSeer [9]	4 B egress per-DL sequence	in-fabric, passive	exact DL	not addressed	next packet, $\Theta(1)$; not measured under spray	exact DL, not spray-measured	no
LinkGuardian [10]	B sequence, era bit, dummy packet	in-fabric, passive	exact DL	not addressed	$\Theta(1)$	exact DL, not spray-measured	link-local retransmit
LossRadar [11]	invertible Bloom filters at both ends	in-fabric, passive digest	exact DL	not addressed	per-packet decode only below about 0.1% retry, $\Theta(1)$	exact DL below 0.1%; counts above	no
UEC LLR [2]	per-link sequence in the MAC	in-fabric, passive	exact DL	in MAC, not measured		exact DL, no spray number	link-layer retry
RFC 6374 [16]	TX/RX counter pairs	in-fabric, passive	if measured per DL	attribution fails	epoch-skew limited (measured here)	none under spray	no
Synchronized snapshot (RFC 6374 with PTP-latched shadow banks)	counter pairs latched on a common epoch	in-fabric, passive	exact DL if per-DL	attribution needs the spray field	matches the witness if latch skew is below about 10 μ s; costs a second register bank, a bank swap, and fabric-wide time (Section X)	exact DL	no
SprayCheck [7]	per-(leaf, spine) RX Z-test, cross-leaf intersection	in-fabric counts, passive	by intersection	yes	0.78 at 10^{-3} , 0.00 at 10^{-4} ; about $\Theta(1/p)$ (measured here)	exact DL at 1–1.5%; pair at $\leq 10^{-3}$ (measured here)	no
FlowPulse [8]	learned per-port load, 1% threshold	in-fabric + model, passive	per-port rule	uplink diluted by senders	fails at $\leq 10^{-3}$ (measured here)	port set or miss; misses uplinks (measured here)	no
dShark [17]	mirrored per-hop trace analysis	host, post hoc	hop via TTL	not a spray study	localizes blackholes; capture drops	hop from trace	no
NetBouncer [18]	active probes, scalar per link	host, active	per-link scalar bound	path pinning breaks	link is one resource	per link, no DL under spray	no
Bartolini et al. [19]	identifiability bound	theory		owns the framing	bound, no cost curve	monitor placement	no
OmniPath Ping [20]	in-network per-hop cache, probes	in-fabric, active	per hop	yes (concurrent)	tracing, not a cost study	per hop	no
CorrOpt [5]	corruption rate, disable, repair	host + counters	per-link scalar	fixed routing	not a spray localizer	no	disable both directions, repair, re-enable quarantine, audit, certify
Aegis [6]	enhanced probe sizes	host, active	path	cannot reach spray-hidden links	64 B probes missed a >1 KB fault	path level	
REPS [3]	spray load balancer, reroute away	in-fabric balancer	not a localizer	yes, avoidance	no loss localization	no	reroute away, re-explore
This paper	2 B egress per-DL sequence, two registers	in-fabric, passive on stamped stream	exact DL	yes	1.00 at 10^{-3} and 10^{-4} , flat 22 M packets	exact DL, set size 1, both families	not claimed

rather than to a directed link. Bartolini et al. [19] give the identifiability bounds of Boolean tomography as a function of routing, the framing in which the aliasing limit of Section II-B sits; the ledger sidesteps the bound by adding information, and we make no identifiability claim.

Repair and restoration. CorrOpt [5] disables a corrupting link in both directions and re-enables it after repair; Aegis [6] quarantines, audits with host-generated traffic, and certifies; REPS [3] reroutes sprayed traffic away from a lossy path and re-explores it. All three re-admit on evidence from real or generated traffic, which a link that spraying has starved cannot supply. Compared to this work, they serve a different objective, the lifecycle after localization, and we cite them to bound what the measurement enables rather than to claim restoration. We have found no prior system that reports the head-to-head of Table IV’s last row against its SprayCheck and FlowPulse rows on a sprayed fabric, on silicon, at the low-loss regime, and that row is what this paper adds.

XII. CONCLUSION

We measured what a two-byte per-directed-link witness buys, and costs, on a packet-sprayed fabric against the two most recent passive detectors built for such fabrics and against

a zero-byte counter pair with identical information, all replayed over one shared stream, and we measured the witness itself on an Intel Tofino switch. A count test and the witness both pay packets that grow as the inverse square of the loss rate, and the witness divides the constant by the background floor. From 1.5 percent loss down to 10^{-4} it detects on one epoch of 2 million packets, where the passive cost reaches 94 million at 10^{-3} and exceeds a 160-million-packet budget at 10^{-4} ; its own wall sits at a few times the floor and moves with it. It names the exact directed link where the passive detectors return the uplink and downlink pair. A counter pair matches it only when both registers are read within about ten microseconds, four orders of magnitude tighter than the census we measured, so what the two bytes buy is in-band alignment rather than information. On silicon the ledger recovers injected loss exactly down to six packets in sixty thousand. It costs 0.14 to 2.4 percent of the wire depending on packet size, one pipeline stage, a rate bound set by the 16-bit sequence and the read loop, and two defects the reduction to two bytes introduced. The advantage widens with several independent faults and fails under a fleet-wide shift and under incast, where a relative passive test is the better tool, so every claim is stated for independent faults on a fabric that otherwise loses at its floor. Our own simulation bounds the completion-time value of

naming a 10^{-3} link at 1.4 percent per collective. The witness is an existing primitive; the contribution is the measurement, its cost, and its boundary. In future work, we intend to design and evaluate a gate that separates shared loss, whether fleet-wide or behind one congested queue, from link loss, so that the ledger's exact localization survives it.

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Philip Akekulip [Biography to be added by the author.]